Joint embedding for finite order forests: a decision theorem and a width classification

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Abstract

Given finitely many forbidden finite order forests, can one decide whether any two permitted forests have a permitted common extension? We answer this question effectively by passing to comparability graphs, decomposing a regular language of binary union/join representations into tree ideals, and deciding inclusion between their graph images. The transfer respects existence of embeddings, although a particular graph embedding need not preserve the order. We also prove an explicit classification: forests of width at most m have the joint embedding property exactly when $m \leq 3$ (with width zero included if the empty forest is allowed). The negative cases have finite certificates, and the positive cases yield countable forests with the prescribed age. The algorithm uses the effective ranked-tree ideal decomposition of Goubault-Larrecq and Schmitz and the effective Parikh theorem. The graph-image, inclusion, order-theoretic and width arguments are given in full.

1 The question and the role of the graph reduction

An order forest is a partially ordered set in which the predecessors of each element form a chain. A finite forest can have several minimal elements. An embedding preserves and reflects the order. In particular, it need not preserve a root, an immediate-predecessor relation, or a meet; none of these is part of the language.

For a finite family $\mathcal{U}$ of finite order forests, put

$$\mathcal{C}(\mathcal{U}) = \{T : T \text{ is a finite order forest and no } U \in \mathcal{U} \text{ embeds into } T\}.$$

A class has the *joint embedding property* (JEP) if any two members embed into a third member. Gonzalez, Rossegger and Turetsky ask which finite forbidden antichains have this property in Question 2 of [4]. Here an antichain of forbidden structures is under the embedding relation; it need not consist of poset antichains.

The question is algorithmic as well as structural. Failure of JEP can be witnessed by two finite forests, but searching for such a pair does not decide the positive case. The missing positive termination guarantee comes from an effective finite ideal cover. Two issues require proof: translating order embeddings into graph embeddings, and recognizing all binary representations of a graph when testing the ideal cover. Related graph JEP decision questions were studied by Carter [1]; the algorithm proved here uses the independent tree-language inputs stated in Section 3.

As a first example, forbidding a two-element antichain leaves just chains. Two finite chains embed into a sufficiently long chain, so JEP holds. Forbidding a five-element antichain instead

allows width four. We shall see that a balanced four-leaf binary tree and a four-leaf comb have no common extension of width four. Thus the obstruction is branching shape, even though the embeddings do not name branching operations.

2 Recovering the order from its comparability graph

Write $\mathrm{Comp}(T)$ for the simple graph on T in which distinct vertices are adjacent exactly when they are comparable. Write P_4 for the four-vertex path and C_4 for the four-vertex cycle. Graph embeddings below are induced embeddings. For a graph family $\mathcal{F}$, $\mathrm{Forb}_{\mathrm{ind}}(\mathcal{F})$ denotes the finite graphs avoiding every member of $\mathcal{F}$ as an induced subgraph.

Lemma 2.1. *The comparability graphs of finite order forests are precisely* $\mathrm{Forb}_{\mathrm{ind}}(P_4, C_4)$. *Each such graph determines its order forest up to isomorphism. Moreover, for finite order forests* S, T,

$$S \hookrightarrow T \quad \Longleftrightarrow \quad \mathrm{Comp}(S) \hookrightarrow \mathrm{Comp}(T).$$

Proof. Each connected component of a finite forest has a least element: the root beneath a vertex is unchanged along any edge of its comparability graph, and hence along a connected component. Its graph is obtained by adding a universal vertex to the disjoint union of the graphs above that root. Recursion excludes induced P_4 and C_4: each forbidden graph is connected and neither has a universal vertex.

Conversely, a nonempty connected graph avoiding P_4, C_4 has a universal vertex. Choose v with maximal size of its closed neighborhood $N[v]$. If v is not universal, a shortest path to a nonneighbor has length two, since a longer shortest path would contain an induced P_4. Call the path v, w, x. Because $x \in N[w] \setminus N[v]$ and $|N[v]| \geq |N[w]|$, choose $u \in N[v] \setminus N[w]$. The four distinct vertices u, v, w, x induce P_4 when ux is absent and C_4 when it is present. Both are impossible. Remove a universal vertex, recursively represent the remaining components, and place the removed vertex below them. Represent a disconnected graph by a disjoint union.

For uniqueness, all universal vertices in a connected representing forest form an initial chain. They are pairwise comparable. If $z \leq u$ and u is universal, then z is comparable with every vertex: use transitivity above u, and linearity of the predecessors of u below it. Thus z is universal too. Every nonuniversal vertex lies above the whole universal chain. Deleting that chain leaves the uniquely specified graph components. Induction reconstructs their orders up to isomorphism. Permuting the names of the universal vertices does not change that isomorphism type.

An order embedding gives an induced graph embedding. For the converse, give the image of a graph embedding the order induced from T. It is a forest whose comparability graph is isomorphic to $\mathrm{Comp}(S)$. Uniqueness gives an order isomorphism from S to that image. □

The last argument asserts existence of an order embedding, not that the original graph map is an order embedding. For a two-element chain, the transposition is a graph automorphism but reverses the order. That distinction is harmless for JEP and essential for interpreting the reduction correctly. These graphs are commonly called trivially perfect graphs; the proof above supplies all properties needed here.

3 Deciding the forbidden-family question

The comparability-graph correspondence reduces the question to a hereditary class of graphs. A graph can have many binary union/join representations, and those representations need not embed into a common tree even when their graphs have a common host. We therefore decompose a language of representations into directed pieces and compare their *graph images*. This section gives the effective steps, including the passage between different binary representations of the same graph.

We use two standard effective results. First, for a regular language of ranked trees, one can compute its homeomorphic downward closure as a finite union of regular ideals. The precise input is the construction of Goubault-Larrecq and Schmitz [2], Section 5: Proposition 18 constructs simple tree regular expressions; Fact 19 and Lemma 20 give a terminating normalization; Theorem 23 identifies the resulting tree products as ideals. Second, the Parikh image of a context-free language has an effective Presburger description. One explicit construction is given by Esparza, Ganty, Kiefer and Luttenberger [3], Theorem 1.1 and Section 3. Presburger arithmetic is first-order arithmetic with addition and order on the natural numbers; its sentences are decidable. These inputs concern tree languages and finite grammars. The graph-containment and JEP transfers below are proved here.

3.1 Binary representations and ideals

Use the ranked alphabet $\{a : 0, U : 2, J : 2\}$. Each occurrence of a represents one vertex. The operations U and J represent disjoint union and join, respectively. Write $\mathsf{E}(t)$ for the graph represented by a term t. Children are ordered in a term, although graph evaluation forgets their order. The empty graph is treated separately.

The ordered homeomorphic relation $\preceq$ on these terms has the following recursive definition. A source can embed into either child of a target; alternatively, equal root symbols can be coupled with childwise embeddings. Thus $a \preceq a$, and for $o \in \{U, J\}$ the coupling clause is

$$o(s_1, s_2) \preceq o(t_1, t_2) \quad \text{if } s_1 \preceq t_1 \text{ and } s_2 \preceq t_2.$$

This is exactly the ranked relation in Example 7 of [2]. It preserves the root operation at each coupled node. An *ideal* is a nonempty downward-closed directed set under the specified relation.

Lemma 3.1. *If $s \preceq t$, then $\mathsf{E}(s)$ embeds as an induced subgraph of $\mathsf{E}(t)$. For every nonempty subset B of the leaves of t, there is a term $t|B \preceq t$ representing exactly $\mathsf{E}(t)[B]$.*

Proof. For the first assertion, diving includes a child graph as an induced subgraph of its parent. Coupling combines two child embeddings on disjoint vertex sets: the common operation puts no edges, or all edges, between those sets on both sides. Induction proves the assertion.

For the second, delete leaves outside B recursively. At a node discard empty child results; return the surviving child if there is only one, and retain the original binary operation if both survive. Suppressing a unary node keeps the surviving child's own operation. Restriction commutes with both graph operations, which proves the graph identity. The dive and coupling clauses prove $t|B \preceq t$ by induction. Adjacent equal operations need not be flattened: the result is a binary representation. □

Let $\mathcal{F}$ be a finite forbidden graph family containing P_4, C_4, and put $\mathcal{K} = \mathrm{Forb}_{\mathrm{ind}}(\mathcal{F})$. Every nonempty member of $\mathcal{K}$ has a binary representation: the construction in Lemma 2.1 recursively takes disjoint unions and adds a universal vertex by joining with a.

Lemma 3.2. *The language $L = \{t : \mathsf{E}(t) \in \mathcal{K}\}$ is effectively regular and downward closed under $\preceq$.*

Proof. Choose $n \geq 1$ at least as large as every forbidden graph. A state records the set of induced graph isomorphism types with at most n vertices present in the represented graph, including the empty type. There are finitely many such types, and canonical representatives and isomorphism tests are finite computations. The leaf state is $\{\varnothing, K_1\}$. For either operation o, the transition on states S, T is

$$\{o(G, H) : G \in S,\ H \in T,\ |G| + |H| \leq n\},$$

with graph types identified up to isomorphism. The empty graph is an identity for both operations. Every induced restriction of an operation splits into restrictions of its two arguments, so these transitions compute the stated profile exactly. Accept states containing no member of $\mathcal{F}$. This constructs the required finite automaton. Heredity and Lemma 3.1 give downward closure. □

Apply the effective ideal-decomposition theorem to obtain

$$L = I_1 \cup \cdots \cup I_r, \tag{1}$$

where the I_i are regular term ideals with effectively available automata. If a forbidden graph is empty, $\mathcal{K}$ is empty and has vacuous JEP. Otherwise it contains the empty graph. If L is empty, $\mathcal{K}$ contains only that graph and has JEP too. These cases are decidable; henceforth $r \geq 1$.

Lemma 3.3. *Put $D_i = \{\varnothing\} \cup \{\mathsf{E}(t) : t \in I_i\}$, on isomorphism types. Each D_i is a hereditary graph class with JEP, their union is $\mathcal{K}$, and*

$$\mathcal{K} \text{ has JEP} \quad \Longleftrightarrow \quad \mathcal{K} = D_i \text{ for some } i. \tag{2}$$

Proof. Containment and covering follow from (1) and the representation of every nonempty graph in $\mathcal{K}$. For heredity take a representation in I_i and apply the pruning part of Lemma 3.1; the result stays in I_i. For JEP take two representing terms in I_i, use its directedness, and evaluate a common term host. Empty graphs cause no difficulty.

Equality with one D_i implies JEP. Conversely, suppose $\mathcal{K}$ has JEP but each D_i is proper. Choose $G_i \in \mathcal{K} \setminus D_i$. Iterating JEP supplies one $H \in \mathcal{K}$ containing all the finitely many G_i. The cover puts H in some D_j; heredity then puts G_j there, a contradiction. □

The remaining task is to decide the finitely many inclusions $\mathcal{K} \subseteq D_i$. Individual binary representations cannot be used for this test, since a graph may have another representation accepted by I_i. We now give the required exact test.

3.2 Recognizing graph images of regular term languages

A *reduced cotree* has leaves a and unordered internal nodes U, J of arity at least two, with adjacent internal operations unequal. Flattening adjacent equal operations turns a binary representation into such a cotree without changing its graph.

Lemma 3.4. *Every graph with a union/join representation has a unique reduced cotree up to isomorphism. All its binary representations are obtained by independently representing each cotree child, then choosing a binary association and a permutation of the children at each operation node.*

Proof. At a reduced U root each child graph is connected: a child is a leaf or has a join root. Its children are therefore precisely the graph's connected components. At a J root the graph is connected and its children are precisely the connected components of the complement. These decompositions recover the root operation and its child vertex sets from the graph. Induction recovers every child cotree; zero and one vertices are immediate. This proves uniqueness. Flattening any binary representation yields this cotree, and reversing each flattening uses precisely binary association and permutation within a single operation. This proves the last assertion. □

Let M be a deterministic complete binary tree automaton with finite state set Q, leaf state q_a, transitions δ_U, δ_J, and accepting set A_M. For a reduced cotree T define

$$S_M(T) = \{q \in Q : \text{some binary representation of } T \text{ has state } q\}. \tag{3}$$

Storing the full subset accounts for all possible representations.

Lemma 3.5. *For each $o \in \{U, J\}$ and $q \in Q$ one can compute a Presburger predicate $P_{o,q}$ on the counts of child state subsets such that, if an o node has children $T_1, \ldots, T_k$, then*

$$q \in S_M(T) \quad \Longleftrightarrow \quad P_{o,q}\big((c_B)_{B \subseteq Q}\big), \qquad c_B = |\{j : S_M(T_j) = B\}|. \tag{4}$$

Equal child occurrences are counted with their full multiplicity.

Proof. Use nonterminals N_q for $q \in Q$ and one terminal b_B for every subset $B \subseteq Q$. For the operation o form the finite grammar

$$N_{\delta_o(r,s)} \longrightarrow N_r N_s \quad (r, s \in Q), \qquad N_q \longrightarrow b_B \quad (q \in B).$$

Compute the Parikh predicate for start nonterminal N_q. A binary representation of the node consists of a binary o skeleton, with each child represented at a chosen state belonging to its subset. Reading that skeleton gives a derivation with the stated terminal counts.

Conversely a derivation with those counts supplies an o skeleton and one selected state for each terminal occurrence. Match occurrences to the actual children of the same subset type. For each occurrence choose a child representation attaining its selected state, and assemble the representations in the skeleton's order. This is a representation of T with root state q. Lemma 3.4 ensures that these are exactly the allowed representations. The effective Parikh theorem therefore proves both the correspondence and effectivity. □

The resulting deterministic cotree automaton has states (ℓ, S), where $\ell \in \{a, U, J\}$ records the root kind and $S \subseteq Q$. Its leaf state is $(a, \{q_a\})$. At an o node, transition to a specified subset S is expressed by

$$\bigwedge_{q \in S} P_{o,q}(c) \ \wedge \ \bigwedge_{q \notin S} \neg P_{o,q}(c). \tag{5}$$

Require at least two children and exclude children with operation o. The counts c_B are sums of the counts for the permitted child root kinds with subset B. These are linear expressions, so the horizontal transitions remain Presburger formulas. Accept when $S \cap A_M$ is nonempty. Induction using (4) shows that this automaton accepts exactly the graphs represented by terms accepted by M, in their canonical cotree form.

Lemma 3.6. *Inclusion of languages recognized by these deterministic cotree automata is decidable.*

Proof. Work throughout on the domain of reduced cotrees. Complement changes the accepting set. For an intersection, use pairs of states; replace each component child-state count by the sum of pair-state counts projecting to it, and conjoin the transition formulas. Root kinds, arity and alternation agree in the product. These are finite Presburger constructions.

For emptiness, start with the actual leaf states and compute a least fixed point of productive states. Given the current finite set R, add an internal state if its transition formula has a nonnegative count vector of total at least two, with count zero on every child type outside R. Presburger decision settles each such question. There are finitely many states, so the process terminates.

Each newly added state has a finite cotree representative: take the required number of copies of representatives for its already productive child states. Conversely induction on cotree height puts every state of an actual finite cotree into the fixed point. Thus a language is nonempty exactly when an accepting state is productive. Apply this to the first language intersected with the complement of the second. □

3.3 The resulting algorithm

For the automata of L and each I_i, construct the cotree automata (3)–(5). They recognize $\mathcal{K}$ and D_i, respectively, with the common empty graph omitted. Lemma 3.6 decides $\mathcal{K} \subseteq D_i$. Since $D_i \subseteq \mathcal{K}$ already, this tests equality. By (2), return yes if some equality holds and no otherwise. Every stage terminates, by the cited effective theorems or the explicit finite-state procedure just proved.

For example, the independent graphs with at most four vertices have JEP. Their binary representations include a four-leaf balanced tree and a four-leaf comb, which cannot have a common binary host with at most four leaves under $\preceq$. Our algorithm decomposes the representation language instead of testing its JEP. In this finite example the maximal binary representations generate principal term ideals. Each such ideal has graph image exactly the independent graphs of sizes one through four, by pruning and leaf counts. Their graph images therefore give the correct affirmative verdict.

Theorem 3.7. *There is an algorithm which, given any finite family $\mathcal{U}$ of finite order forests, decides whether $\mathcal{C}(\mathcal{U})$ has JEP. More precisely, set*

$$\mathcal{F} = \{P_4, C_4\} \cup \{\mathrm{Comp}(U) : U \in \mathcal{U}\}.$$

Then $\mathcal{C}(\mathcal{U})$ has JEP if and only if $\mathrm{Forb}_{\mathrm{ind}}(\mathcal{F})$ has JEP.

Proof. Lemma 2.1 gives an effective correspondence on isomorphism types between finite forests and $\mathrm{Forb}_{\mathrm{ind}}(P_4, C_4)$, preserving and reflecting existence of embeddings. Thus avoiding $\mathcal{U}$ corresponds exactly to avoiding the displayed graph family, and common hosts correspond in both directions. Compute this finite graph list and run the terminating graph procedure established above. □

An input family need not be an antichain: remove redundant forbidden members by finite embedding tests. The algorithm supplies an effective criterion, with no claim that its large intermediate automata are practical or that the decision procedure has been implemented.

Proposition 3.8 (Finite certificates for negative instances)**.** *If $\mathcal{C}(\mathcal{U})$ fails JEP, one can compute a pair F, G in the class and a finite exhaustive certificate that they have no common extension in the class.*

Proof. Any common host can be replaced by its induced suborder on the union of the two embedding images. Heredity preserves membership in the class, and the resulting host has at most $|F| + |G|$ vertices. For a fixed pair it therefore suffices to enumerate all forests of size at most that bound and test their membership and all candidate embeddings. These are finite calculations on finite order relations.

After a negative verdict from Theorem 3.7, enumerate permitted pairs and perform this finite test for each. A bad pair must eventually appear. The pair together with the exhaustive host test is a certificate verifiable by finite host enumeration alone. □

The bound is a bound on the size of a host for a fixed pair. It is not a uniform bound on the size of the first bad pair as a function of the forbidden family. Without the total decision procedure the same search would semidecide only failure of JEP.

4 A sharp classification by width

The width of a finite poset is the largest size of an antichain. For a finite forest it is the number of maximal elements, which we call leaves. Indeed, each element of an antichain extends to a leaf, and incomparable elements have disjoint upper cones: a common upper bound would have incomparable predecessors.

Theorem 4.1. *For $m \geq 1$, the class of finite order forests of width at most m has JEP if and only if $m \leq 3$.*

Proof. First suppose $m \leq 3$. A forest of width at most m embeds as an induced suborder of a rooted binary tree with exactly m leaves. Add a root below all components; replace any multifurcation by binary branching through inserted vertices; and, if necessary, attach extra leaves. These operations preserve the old order and need at most m leaves. Suppressing unary vertices now gives a full binary tree. For each of $m = 1, 2, 3$ there is exactly one unordered full binary tree with m leaves. A sufficiently long subdivision of each edge of this tree, together with a sufficiently long initial stem, therefore contains any prescribed pair of the original forests. Choose on each segment enough vertices for both individual embeddings; they need not have disjoint images. This common host still has width m.

Let $m \geq 4$. Compare a binary comb with m leaves, whose root split is $1 + (m-1)$, with a full binary tree whose root split is $2 + (m-2)$. For a vertex v let $L(v)$ be its set of descendant leaves. Distinct leaf clusters form a laminar family: two are nested or disjoint.

Suppose a full binary m-leaf tree embeds into a host H of width at most m. Its leaf images are incomparable; choose a host leaf above each one. There must be exactly m host leaves, and each source-leaf image has only its chosen host leaf above it. For a source vertex v, the host leaves above its image are precisely the matched leaves in $L(v)$. To check the nontrivial inclusion, any host leaf is above the image of a unique source leaf ℓ. If it is also above the image of v, foresthood makes those two images comparable. Order reflection then makes v and ℓ comparable, so $v \leq \ell$.

A full binary tree has $2m - 1$ distinct nonempty clusters. This family is maximal laminar on the m leaves. Indeed, a set compatible with both children of every binary split must either be contained in one child or contain that entire child; descending recursively shows that it is already a cluster. The host's cluster family contains the source's and is laminar, so the two families coincide. Thus a common host for the two chosen sources would identify their cluster families under a leaf bijection. In the comb all nonsingleton proper clusters are nested. The other source has two disjoint nonsingleton root-child clusters. This is impossible. □

If A_k is the k-element poset antichain, then

$$\mathcal{C}(\{A_k\}) \text{ has JEP} \quad \Longleftrightarrow \quad k \leq 4 \qquad (k \geq 2).$$

For $m = 4$ the two negative witnesses each have seven vertices: the balanced binary tree and the four-leaf comb. Their embeddings were not assumed to preserve roots or meets. Cluster preservation was derived from the saturated width bound. If empty forests are allowed, the width-zero class consists only of the empty forest and has JEP too.

5 Countable realizations and scope

The age of a countable forest is the class of isomorphism types of its finite induced suborders. The following standard age construction explains one use of JEP without imposing amalgamation or homogeneity.

Proposition 5.1. *A nonempty hereditary class $\mathcal{K}$ of finite forests is the age of a finite or countable forest if and only if it has JEP. If it contains forests of unbounded size, the realizing forest is countably infinite.*

Proof. Any two finite suborders of one forest are contained in their finite union, proving necessity. Conversely enumerate the members of $\mathcal{K}$ up to isomorphism. Use JEP repeatedly to obtain a chain of members, with each enumerated member embedding into a later stage. Rename each successive host so that its chosen copy of the preceding stage is an actual induced suborder. The union is a forest: every finite violation of the order or predecessor-chain axioms would already occur at a finite stage. Every finite suborder of the union is contained in a stage, and hence belongs to $\mathcal{K}$ by heredity. Every member of $\mathcal{K}$ embeds into some stage. The union therefore has exactly the required age. Unbounded size forces the union to be infinite. □

Together with Theorem 3.7, this decides which finite forbidden-family forest classes are ages. The empty class is treated separately: under the usual vacuous convention it has JEP, but is the age of no structure. Whether the empty forest is admitted as a structure is another convention that can be settled by a finite input check.

The source motivation in [4] concerns Scott analysis below the Vaught ordinal. The present paper answers its finite-forest Question 2 by an effective criterion, and gives a direct structural answer for antichain obstructions. It does not establish a model-counting theorem for arbitrary complete first-order theories. Nor does the age construction alone assign a Scott rank to its union. Such conclusions require the additional hypotheses of the relevant results in [4].

Preparation and status. This manuscript was assembled with AI assistance from a research programme directed by the repository owner. The effective ideal-decomposition and Parikh theorems are credited as published inputs. The graph-image and inclusion lemmas, forest transfer and width proofs above are offered for review; no implementation or formal verification of the full decision algorithm is claimed. No historical priority claim is made for those deductions. The repository contains the full manuscript and its build instructions; it requires no private project notes or conventions.

References

[1] D. Carter, *On the joint embedding property for cographs and trees*, arXiv:2409.06127v1 (2024). https://arxiv.org/abs/2409.06127v1.

[2] J. Goubault-Larrecq and S. Schmitz, *Deciding Piecewise Testable Separability for Regular Tree Languages*, ICALP 2016, LIPIcs 55, 97:1–97:15, Example 7 and Section 5. https://doi.org/10.4230/LIPIcs.ICALP.2016.97.

[3] J. Esparza, P. Ganty, S. Kiefer and M. Luttenberger, *Parikh's Theorem: A simple and direct automaton construction*, arXiv:1006.3825v3 (2011), Theorem 1.1 and Section 3. https://arxiv.org/abs/1006.3825v3.

[4] D. Gonzalez, D. Rossegger and D. Turetsky, *Scott Analysis below the Vaught Ordinal*, arXiv:2606.15205v2 (2026), Question 2. https://arxiv.org/abs/2606.15205v2.